1 Family of Curves

Recall: Recall that an **admissible curve** $\gamma : I \to M$ is a piecewise smooth curve for a partition $(a_0, a_1, \dots, a_n)$ of I.

Recall: For an admissible curve $\gamma : [a, b] \to M$, where (M, g) is a Riemannian manifold, we define its **length** $L_g(\gamma)$ by

$$L_g(\gamma) := \int_a^b |\gamma'(t)|_g dt,$$

where $|v|_g := \sqrt{g_p(v, v)} \triangleq \langle v, v\rangle^{1/2}$ for all $v \in T_pM$.

Recall: Let (M, g) be a Riemannian manifold. For each pair of points $p, q \in M$, the **Riemannian distance** from p to q is defined to be

$$d_g(p, q) := \inf\{L_g(\gamma)|\ \gamma \text{ is an admissible curve from } p \text{ to } q\}.$$

Minimizing Curves

Definition 1.1. : Let (M, g) be a Riemannian manifold. An admissible curve γ in M is said to be a **minimizing curve** if $L_g(\gamma) \le L_g(\tilde{\gamma})$ for every admissible curve $\tilde{\gamma}$ with the same endpoints.

Remark: It's clear that if M is connected, then γ is minimizing $\Leftrightarrow$ $L_g(\gamma)$ is the Riemannian distance between its endpoints.

Definition 1.2. : Let $I, J \subseteq \mathbb{R}$ be two intervals, a continuous map

$$\Gamma : I \times J \to M$$

is called a **one-parameter family of curves**. Such family defines two collections of curves in M:

(1) the **main curves** $\Gamma_s(t) := \Gamma(s, t)$ defined for $t \in I$ with s constant;

(2) the **transverse curves** $\Gamma^{(t)}(s) := \Gamma(s, t)$ defined for $s \in J$ with t constant.

Admissible Family of Curves

Definition 1.3. : A one-parameter family of curves Γ is called an **admissible family of curves** if

(1) its domain is of the form $J \times [a, b]$ for some open interval J;

(2) there is a partition $(a_0, a_1, \dots, a_n)$ of $[a, b]$ such that Γ is smooth on each rectangle of the form $J \times [a_{i-1}, a_i]$;

(3) $\Gamma_s(t)$ is an admissible curve for each $s \in J$.

Remark: Let Γ be as above, we denote

$$\partial_t\Gamma(s,t) := (\Gamma_s)'(t), \quad \partial_s\Gamma(s,t) := (\Gamma^{(t)})'(s).$$

Variation of Smooth Curves

Definition 1.4. : For a given admissible curve $\gamma : [a,b] \to M$, a **varation of** γ is an admissible family of curves $\Gamma : J \times [a,b] \to M$ such that J is an open interval containing 0 and $\Gamma_0 = \gamma$. Additionally, Γ is called a **propoer variation** if $\Gamma_s(a) = \gamma(a), \Gamma_s(b) = \gamma(b)$ for all $s \in J$.

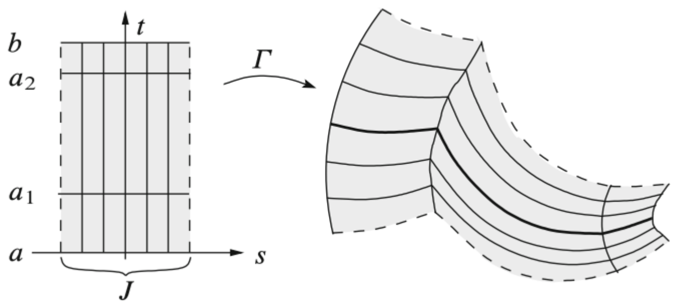

图 1: An admissible curves

Variation Field

Definition 1.5. : If Γ is a variation of the curve $\gamma : [a,b] \to M$, the **variation field of** Γ is the piecewise smooth vector field $V(t) := \partial_s\Gamma(0,t)$ along γ. We say a vector field V along γ is **proper** if $V(a) = 0$ and $V(b) = 0$.

Recall: The **expoential map** is defined by

$$\exp : \varepsilon \to M$$

$$v \mapsto \gamma_v(1)$$

where $\varepsilon := \{v \in TM | \ \gamma_v$ is defined on an interval containing $[0,1]\}$ and γ_v is the unique maximal geodesic with $(\gamma_v)'(0) = v$. For each $p \in M$, denote $\varepsilon_p := T_pM \cap \varepsilon$ and $\exp_p := \exp|_{T_pM} : \varepsilon_p \to M$, $v \mapsto \gamma_v(1)$ with $\gamma_v(0) = p, (\gamma_v)'(0) = v$.

Recall: (1) For each $v \in TM$, the geodesic γ_v is given by $\gamma_v(t) = \exp(tv)$.
(2) For each $p \in M$, the differential map

$$d(\exp_p)_0 : T_0(T_pM) \cong T_pM \to T_pM$$

is precisely the identity map on T_pM, i.e., $d(\exp_p)_0 = \mathrm{Id}_{T_pM}$.

Lemma 1.1. : *If $\gamma : [a,b] \to M$ is an admissible curve and V is a piecewise smooth vector field along γ, then V is the variation field of some variation of γ. If V is proper, the variation can be chosen to be proper.*

Proof: Set

$$\Gamma(s,t) := \exp_{\gamma(t)}(sV(t)),$$

hence

$$\Gamma(0,t) = \exp_{\gamma(t)}(0) = \gamma(t)$$

and

$$\partial_s \Gamma(0,t) = \frac{\partial}{\partial s}\bigg|_{s=0} \exp_{\gamma(t)}(sV(t)) = d(\exp_{\gamma(t)})_0(V(t)) = V(t)$$

thus $V(t)$ is a variation field of Γ, where Γ is a variation of γ.

Moreover, if $V(a) = 0$ and $V(b) = 0$, then $\Gamma_s(a) \equiv \gamma(a)$ and $\Gamma_s(b) \equiv \gamma(b)$. $\square$

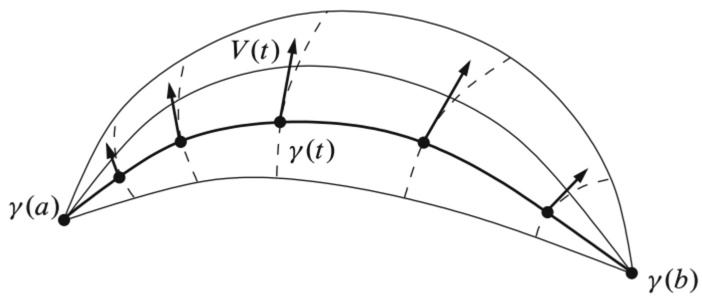

图 2: Every vector field along γ is a variation field

Remark: If V is a piecewise smooth curve along Γ, i.e., V is a vector field along the admissible family Γ whose restriction on each rectangle $J \times [a_{i-1}, a_i]$ is smooth for some partition $(a_0, a_1, \ldots, a_k)$ for $[a,b]$. Then we can compute the covariant derrivative of V along the main curves or transverse curves (with respect to the Levi-Civita connection), which is denoted by $D_t V$ and $D_s V$ repectively.

Recall: Let $\gamma : I \to M$ be a smooth curve in M, ∇ is a connection on TM, V is a smooth vector field along γ. Take a local coordinate $(U; x^i)$ for $\gamma(t)$ and write V, γ as $V(t) \triangleq V^j(t)\partial_j|_{\gamma(t)}$, $\gamma(t) \triangleq (\gamma^1(t), \gamma^2, \ldots, \gamma^n(t))$, then the covariant derivative of V along γ w.r.t. ∇ (denoted by $D_t V$) is given by

$$D_t V(t) = (\dot{V}^k(t) + \dot{\gamma}(t) V^j(t) \Gamma_{ij}^k(\gamma(t)))\partial_k|_{\gamma(t)}.$$

Here, Γ_{ij}^k is the Christoffel symbol that is given by $\nabla_{\partial_i}\partial_j := \Gamma_{ij}^k \partial_k$.

Symmetry Lemma

Lemma 1.2. : *Let $\Gamma : J \times [a,b] \to M$ be an admissible family of curves in a Riemannian manifold. On every rectangle $J \times [a_{i-1}, a_i]$ where J is smooth, we have*

$$D_s \partial_t \Gamma = D_t \partial_s \Gamma.$$

Proof: For any point $\Gamma(s_0, t_0)$, take a local coordinate (x^i) around it. Denote

$$\Gamma(s,t) \triangleq (x^1(s,t), x^2(s,t), \ldots, x^n(s,t)).$$

Then we get

$$\partial_t\Gamma = \frac{\partial x^k}{\partial t}\partial_k, \quad \partial_s\Gamma = \frac{\partial x^k}{\partial s}\partial_k.$$

Thus we have

$$D_s\partial_t\Gamma = (\frac{\partial^2 x^k}{\partial s\partial t} + \frac{\partial x^i}{\partial t}\frac{\partial x^j}{\partial s}\Gamma_{ij}^k)\partial_k,$$
$$D_t\partial_s\Gamma = (\frac{\partial^2 x^k}{\partial t\partial s} + \frac{\partial x^i}{\partial s}\frac{\partial x^j}{\partial t}\Gamma_{ji}^k)\partial_k.$$

Since the Levi-Civita connection is symmetry, i.e., $\Gamma_{ij}^k = \Gamma_{ji}^k$, the conclusion holds. □

2 Minimizing Curves Are Geodesics

Recall: For a Riemannian manifold (M, g) and ∇ a connection on TM, if V, W are two smooth vector fields along the curve γ, then we have

$$\frac{d}{dt}\langle V, W\rangle = \langle D_tV, W\rangle + \langle V, D_tW\rangle.$$

First Variation Formula

Theorem 2.1. *: Let (M, g) be a Riemannian manifold. Suppose $\gamma : [a, b] \to M$ is a unit-speed curve admissible curve, $\Gamma : J \times [a, b] \to M$ is variation of γ, and V is its variation field. Then $L_g(\Gamma_s)$ is a smooth function of s, and*

$$\left.\frac{d}{ds}\right|_{s=0} L_g(\Gamma_s) = -\int_a^b \langle V, D_t\gamma'\rangle dt - \sum_{i=1}^{k-1}\langle V(a_i), \Delta_i\gamma'\rangle + \langle V(b), \gamma'(b)\rangle - \langle V(a), \gamma'(a)\rangle,$$

where $(a_0, a_1, \ldots, a_k)$ is an admissible partition for V and for each $i = 1, 2, \ldots, k-1$, $\Delta_i\gamma' = \gamma'(a_i^+) - \gamma'(a_i^-)$ is the jump in the velocity vector field γ' at a_i. In particular, if Γ is a proper variation, then

$$\left.\frac{d}{ds}\right|_{s=0} L_g(\Gamma_s) = -\int_a^b \langle V, D_t\gamma'\rangle dt - \sum_{i=1}^{k-1}\langle V(a_i), \Delta_i\gamma'\rangle.$$

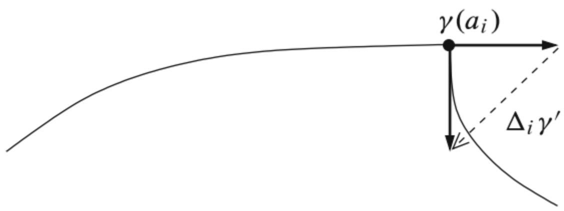

图 3: $\Delta_i\gamma'$ is the jump in γ' at a_i

Proof: It's easy to see that $L_g(\Gamma_s)$ is a smooth function of s by the knowledge of calculus.

For simplicity, denote

$$T(s,t) := \partial_t\Gamma(s,t), \quad S(s,t) := \partial_s\Gamma(s,t)$$

By differentiating $L_g(\Gamma_s)$ on the interval $[a_{i-1}, a_i]$, we get

$$\frac{d}{ds}L_g(\Gamma_s|_{[a_{i-1},a_i]}) = \int_{a_{i-1}}^{a_i} \frac{\partial}{\partial s}\langle T, T\rangle^{1/2} dt = \int_{a_{i-1}}^{a_i} \frac{1}{2}\langle T, T\rangle^{-1/2} 2\langle D_s T, T\rangle dt = \int_{a_{i-1}}^{a_i} \frac{1}{|T|}\langle D_t S, T\rangle dt,$$

Let $s = 0$, note that $S(0,t) = V(t)$ and $T(0,t) = \gamma'(t)$, we get

$$\begin{aligned}\left.\frac{d}{ds}\right|_{s=0} L_g(\Gamma|_{[a_{i-1},a_i]}) &= \int_{a_{i-1}}^{a_i} \langle D_t V, \gamma'\rangle dt = \int_{a_{i-1}}^{a_i} (\frac{d}{dt}\langle V, \gamma'\rangle - \langle V, D_t\gamma'\rangle) dt \\ &= \langle V(a_i), \gamma'(a_i^-)\rangle - \langle V(a_{i-1}), \gamma'(a_{i-1}^+)\rangle - \int_{a_{i-1}}^{a_i} \langle V, D_t\gamma'\rangle dt.\end{aligned}$$

Summing over i, we get the conclusion. $\square$

Remark: The requirement in the above theorem that γ is of unit-speed is not a real restriction but just a computatoinal convenience, since every admissible curve bas a unit-speed reparanetrization and the length of γ is independent of the choice of the parametrization.

Theorem 2.2. *: In a Riemannain manifold, every minimizing curve is a geodesic when it is given by a unit-parametrization.*

Proof: Let $\gamma : [a,b] \to M$ be a minimizing curve of unit-speed, let $(a_0, a_1, \ldots, a_k)$ be an admissible partition for γ. If Γ is a proper variation of γ, then $L_g(\Gamma_s)$ is a smooth function of s that achieves its minimum at $s = 0$, so we must have $\frac{d}{ds}|_{s=0}L_g(\Gamma_s) = 0$. By Lemma 1.1, since every proper vector field along γ is the variation field of some peoper variation of γ, then by the First Variation Formula, we have

$$\left.\frac{d}{ds}\right|_{s=0} L_g(\Gamma_s) = -\int_a^b \langle V, D_t\gamma'\rangle dt - \sum_{i=1}^{k-1}\langle V(a_i), \Delta_i\gamma'\rangle = 0.$$

On each $[a_{i-1}, a_i]$, we prove that $D_t\gamma' = 0$:

Take a bump function $\varphi \in C^\infty(\mathbb{R})$ such that supp$\varphi \subseteq (a_{i-1}, a_i)$, by First Variation Formula with $V := \varphi D_t\gamma'$, we get

$$-\int_{a_{i-1}}^{a_i} \varphi |D_t\gamma'| dt = 0,$$

since the integrand is non-negative and $\varphi > 0$ on (a_{i-1}, a_i), we can know that $D_t\gamma' \equiv 0$ on each $[a_{i-1}, a_i]$.

Next, we show that $\Delta_i\gamma' = 0$ for each i between 0 and k, i.e., γ has no corners. For each i, we can choose a bump function in a coordinate chart to construct a piecewise smooth vector field V along γ such that $V(a_i) = \Delta_i\gamma'$ and $V(a_j) = 0$ for any $j \neq i$. Again, by First Variation Formula, we get

$$-\langle V(a_i), \Delta_i\gamma'\rangle = -|\Delta_i\gamma'|^2 = 0,$$

which implies that $\Delta_i\gamma' = 0$ for each i.

It's clear to see that γ is smooth by the uniqueness of geodesics. □

Actually, the First Variation Formula tells us more than is claimed in Theorem 2.1. In Proving that γ is a geodesic, we did not use the full strength of the assumption that the length of Γ_s takes a minium when $s = 0$, we only used the fact that its derivative is zero.

Generally, we say that an admissible curve γ is a **critical point of** L_g if for every proper variation Γ_s of γ, the derivative of $L_g(\Gamma_s)$ w.r.t. s is zero at $s = 0$.

Corollary 2.1. : *A unit-speed admissible curve γ is a critical point for $L_g \iff$ it is a geodesic.*

Proof: With the same method in the proof in Theorem 2.2, we can show that γ is a geodesic.

Conversely, if γ is a geodesic, then the first term in the right-hand side of the first variation formula

$$\left.\frac{d}{ds}\right|_{s=0} L_g(\Gamma_s) = -\int_a^b \langle V, D_t\gamma'\rangle dt - \sum_{i=1}^{k-1} \langle V(a_i), \Delta_i\gamma'\rangle$$

vanishes by the geodesic equation $D_t\gamma' \equiv 0$ and the second term also vanishes since γ' has no jumps. □

Remark: By the corollary above, we know that the geodesic equation $D_t\gamma' = 0$ characterizes the critical points of the length functional L_g. In general, the equation that characterizes the critical points of a functional on a space of maps is called the **variational equation** or the **Euler-Lagrange equation** of the functional.